A short-bond rule for the isotope effect on polymorph stability in hydrogen-bonded crystals

Muhammet Can Erganiş[1, *]

[1] *Department of Metallurgical and Materials Engineering, Middle East Technical University, Ankara, Türkiye*

(Dated: Revised draft, 23 October 2026)

Deuteration can change which polymorph of a hydrogen-bonded crystal is stable, but a logged literature search found no cross-compound statement of its direction. For 220 carbon-free polymorph pairs of hydrogen compounds we computed the harmonic isotope double difference $\Delta\Delta F_{\mathrm{iso}}$, free of the static energy gap, with three universal machine-learned potentials. At 300 K $\Delta\Delta F_{\mathrm{iso}}$ rises by 0.11 to 0.13 meV per H per mÅ of mean X-H length difference (formula-clustered intervals 0.07 to 0.17), a stretch term half cancelled by bend and libration. Where one polymorph has mean O-H of 0.99 Å or more (a post hoc boundary), deuteration favours the polymorph with the shorter O-H bonds in 74 of 84 pairs and 27 of 28 formulas by model majority (185 of 212 evaluations; constant sign 131); the 10 wrong pairs share one $Al(OH)_3$ structure. The rule holds where the effect exceeds the model spread, has little skill near balance and reverses for weak hydrogen bonds (18 of 20). A one-mode harmonic estimate within the same models explains the stretch slope. A prospective test on new pairs failed. Exploratory quasi-harmonic calculations put the AlOOD diaspore to δ transition about 1 GPa above AlOOH. D/H partitioning between coexisting polymorphs would test it.

I. INTRODUCTION

Deuterium replaces hydrogen in many experiments on crystals. Neutron crystallographers deuterate samples to reduce the incoherent background of hydrogen and often read the deuterated phase as a stand-in for the hydrogenous one. Isotope tracing, isotope geochemistry and synthesis in D_2O or D_2 rely on the same substitution. Whether the swap changes which crystal structure is stable is a practical question, and in some systems the answer is yes. In δ-AlOOH, a candidate carrier of water into the deep Earth, neutron diffraction places the proton-disorder transition from $P2_1nm$ to $Pnnm$ at 9.0 GPa for AlOOH and at 13.4 GPa for AlOOD, a shift of 4.4 GPa, while hydrogen-bond symmetrization is observed at 18.1 GPa in AlOOH only [1, 2]. In KH_2PO_4 deuteration nearly doubles the ferroelectric transition temperature, the canonical solid-state isotope effect on a phase transition [3, 4]. In molecular crystals deuteration selects a different polymorph of pyridine [5] and of the pentachlorophenol and 4-methylpyridine complex [6–8], selects one of two concomitant polymorphs of ROY [9], gives a new phase in a hydrogen-bonded host-guest crystal through a geometric isotope effect [10], and changes the phase transition of 3,5-pyridinedicarboxylic acid [11]; deuterium can perturb molecular arrangement in the solid state more generally [12]. Some of these cases concern which form crystallises, not an equilibrium free-energy difference; the quantity computed here is the latter.

The physics behind these effects is understood in outline. The isotope effect on a free energy comes from the zero-point and thermal energies of the modes in which hydrogen moves [13, 14]. A stronger hydrogen bond lengthens and softens the covalent O-H bond, which lowers the stretch frequency (that longer bonds have smaller force constants is Badger's rule [15]), and stiffens the bending and librational modes. The two changes act in opposite directions on the H/D free-energy difference. McKenzie and co-workers derived this competition for the isotope fractionation and geometry of hydrogen bonds [16, 17], Li, Walker and Michaelides showed that the net quantum effect on hydrogen-bond strength changes sign with the strength of the bond [18], and Markland and Berne showed the same competition in the isotope fractionation of water [19]. Hydrogen bonding is known qualitatively to control D/H fractionation between OH-bearing minerals and water [20, 21], and geometric H/D isotope effects on hydrogen-bond length are well documented [3, 22]. Path-integral methods give isotope effects beyond the harmonic approximation for single systems [23–27], and anharmonic free energies change polymorph rankings in molecular crystals [28–30]. Large-scale harmonic isotope thermodynamics has been reported for 115 metal hydrides [31], where the gas dominates the formation reaction, and universal machine-learned interatomic potentials (MLIPs) now give phonons across many systems [32, 33], without isotopes.

What is not available is a cross-compound, quantitative statement for solid-solid polymorph pairs: given two polymorphs of a hydrogen-bonded compound, which one does deuteration favour, by how much, and where does the statement fail? A logged literature search for this work (20 prior-art queries before submission of the screen, then two rounds of about 55 and about 150 Valency Bond and web-search queries after it, mostly over abstracts and search snippets) found no paper that states such a rule across many compounds or reports a slope of the poly-

* https://orcid.org/0009-0001-1397-6366

morph isotope free-energy difference against O-H length. The statement is bounded by that search. The direction itself is expected from the work cited above; the contribution here is its quantification and its limits across inorganic polymorph pairs.

Isotope effects on polymorph free energies are a few meV per atom, smaller than the static energy errors of fast models. We therefore compute only the isotope double difference, in which the static energy gap between the two polymorphs cancels, with three universal MLIPs over polymorph pairs from the Materials Project and JARVIS-DFT. This paper reports: (i) the net slope of the harmonic H/D free-energy difference between polymorphs against the change in mean X-H length, with clustered uncertainties and a mode-resolved split into stretch and bend and libration terms; (ii) a sign rule derived from it, with pair and formula counts, competitor descriptors, an orientation-invariant null and the one structure in which it fails; (iii) where the rule is reliable, namely where the effect is larger than the model spread, and the reversed sign for weak hydrogen bonds; (iv) an analytic one-mode estimate of the stretch slope within the same models, and the temperature dependence of the rule; (v) every blind and prospective test we ran, including four that failed or did not reach their bar; (vi) an exploratory prediction for the diaspore to δ-AlOOH transition and the measurements that would test the rule. The calculations are harmonic or quasi-harmonic, the potentials share training data, and no polymorph isotope free energy has been measured. The result is a computational relationship within these models, and we state its limits where they arise.

II. METHODS

A. Isotope double difference

For polymorphs a and b of the same composition we compute

$$\Delta\Delta F_{\rm iso}(T) = [F_{\rm D}(b) - F_{\rm H}(b)] - [F_{\rm D}(a) - F_{\rm H}(a)]\,, \quad (1)$$

per H of a common cell unless stated otherwise (meV/atom is used only for the AlOOH card and the code checks, and is labelled). A positive $\Delta\Delta F_{\rm iso}$ means that deuteration destabilises b relative to a. The static energies do not depend on mass and cancel, so errors of the potentials in the polymorph energy gap, often larger than the isotope signal, do not enter. The cancellation does not remove errors in the force constants or the relaxed geometries, or the omitted anharmonicity. Within the Born-Oppenheimer approximation the force constants Φ are mass independent, so one finite-displacement phonon calculation per structure gives both isotopes: the dynamical matrix $D_X = M_X^{-1/2}\Phi M_X^{-1/2}$ is rebuilt with the D mass on every H site and diagonalised again ($m_{\rm H} = 1.00782503$ u, $m_{\rm D} = 2.01410178$ u). For $X \in \{\mathrm{H}, \mathrm{D}\}$,

$$F_X(T) = \sum_{q,\nu} w_q \left[\frac{\hbar\omega^X_{q\nu}}{2} + k_{\rm B}T \ln\left(1 - e^{-\hbar\omega^X_{q\nu}/k_{\rm B}T}\right)\right], \quad (2)$$

with q-point weights w_q summing to one; in this direct sum modes below 0.1 THz, including imaginary modes, are dropped. The classical momentum term $-\frac{3}{2}N_{\rm H}k_{\rm B}T\ln(m_{\rm D}/m_{\rm H})$ is the same for both polymorphs and cancels; at high temperature the surviving term is the Wigner-Kirkwood correction $(\hbar^2/24k_{\rm B}T)(1/m_{\rm D} - 1/m_{\rm H})\sum_i[\mathrm{Tr}\,\Phi^b_{ii} - \mathrm{Tr}\,\Phi^a_{ii}]$ over H sites i, so $\Delta\Delta F_{\rm iso}$ decays as $1/T$ and D prefers the phase with the larger H force-constant trace [13, 14]. An absolute transition pressure, unlike $\Delta\Delta F_{\rm iso}$, depends on the static energy surface; for the pressure predictions of Sec. III H each isotope has its own quasi-harmonic minimization, $G_X(T,P) = \min_V[E(V) + F_{{\rm vib},X}(V,T) + PV]$, reported separately from the harmonic screen. All formulas used in the code were derived twice, independently, and cross-checked numerically before use; derivations and test cases are in the repository [34] as notes M01 to M03.

B. Mode-resolved decomposition

To separate the stretch from the bend and libration we use the exact mass identity obtained from the Hellmann-Feynman theorem applied to $\Phi u = \omega^2 M u$,

$$F_{\rm D} - F_{\rm H} = -\int_{\ln m_{\rm H}}^{\ln m_{\rm D}} \sum_{q,\nu} w_q\, w^{\rm H}_{q\nu}\, \frac{\hbar\omega_{q\nu}}{4} \coth\frac{\hbar\omega_{q\nu}}{2k_{\rm B}T}\, d\ln m, \quad (3)$$

where $e_{q\nu}$ is the normalised mass-weighted eigenvector at H mass m and $w^{\rm H}_{q\nu} = \sum_{i\in{\rm H}}|e_{i,q\nu}|^2$. We split $w^{\rm H} = w^{\rm str} + w^{\perp}$, with $w^{\rm str} = \sum_{i\in{\rm H}}|\hat{b}_i \cdot e_{i,q\nu}|^2$ and $\hat{b}_i$ the unit vector from H to its nearest non-H atom. The integral is evaluated with six-point Gauss-Legendre quadrature in $\ln m$, rediagonalising at each intermediate mass and using the eigenvectors at that mass. This gives $\Delta\Delta F_{\rm iso} = \Delta\Delta F_{\rm str} + \Delta\Delta F_{\perp}$ and the compensation $C = -\Delta\Delta F_{\perp}/\Delta\Delta F_{\rm str}$, which is ill conditioned when the stretch term is near zero. We call the perpendicular term bend and libration; it is an operational component, and crystal modes mix local and collective motion. The projection avoids frequency cuts, which are ill defined because a strongly hydrogen-bonded O-H stretch and its O-D counterpart can fall on different sides of any cut. The quadrature integrates $|\omega|$ for imaginary modes.

C. Potentials and phonons

Every quantitative result comes from three universal MLIPs: MACE (checkpoint `medium-mpa-0`, mace-torch 0.3.16) [35], ORB (`orb_v3_conservative_inf_mpa`, orb-models 0.5.5) [36]

and MatterSim (`MatterSim-v1.0.0-5M`, mattersim 1.2.5) [37]. SevenNet (`7net-mf-ompa`, modal `mpa`, sevenn 0.13.0) [38, 39] enters only as sign votes and in pipeline counts, which are labelled. It was set aside before the screen because its transverse optical H frequency in LiH is 15.3% above experiment (20.41 against 17.7 THz); MACE is 12.8% above (19.97 THz), so the cut separates models at one tolerance rather than on a physical criterion. The potentials are separately trained on overlapping PBE-level data, so their agreement is not an independent validation, and the model spread is a range, not a confidence interval.

Structures were relaxed with each potential, cell included, to a maximum force of 10^{-4} eV/Å with a limit of 500 steps; non-convergence is flagged, not excluded. Force constants were computed with phonopy [40, 41] by finite displacements of 0.01 Å in supercells with a minimum image length of 9.5 Å and at most 160 atoms. Free energies were recomputed on a shifted, Γ-free q mesh with about $60/a$ points along each axis of lattice length a (in Å), rounded to an even number between 4 and 16, which removes the treatment of the acoustic modes at Γ as a source of spurious isotope terms; mesh convergence of $\Delta\Delta F_{\mathrm{iso}}$ was not tested across the screen, and for one card pair (goethite) a mesh change moved a pressure-dependent free energy by 0.1 to 0.2 meV/atom. A structure-model result is called clean when the quadrature of Eq. (3) and the direct difference from Eq. (2) agree within 0.01 meV/atom. Because the two routes treat soft modes differently, the criterion tests the numerical consistency of the isotope term; it does not certify relaxation convergence, dynamical stability or insensitivity to the treatment of imaginary modes. Section III C reports the subset that also has converged relaxations and no imaginary modes.

As a per-structure benchmark against PBEsol phonons of 40 entries of the MDR phonon database [33, 41] (36 for SevenNet), at the DFT geometries, the isotope zero-point energy difference has a mean absolute error of 0.7 to 1.5 meV per H (ORB 0.7, SevenNet 1.1, MatterSim 1.3, MACE 1.5), the 300 K H/D free-energy difference 0.5 to 1.1 meV per H and the X-H stretch frequency 28 to 45 cm^{-1}. These errors are comparable to the median $|\Delta\Delta F_{\mathrm{iso}}|$ of 2.4 meV per H on the headline rows; no DFT value of a polymorph double difference was computed. Quasi-harmonic volume isotope effects, $100(V_{\mathrm{H}}/V_{\mathrm{D}} - 1)$, were compared with measurements before the screen: with the three models brucite $Mg(OH)_2$ gives +0.19 to +0.33% at 290 K against a measured +0.31% [42], LiH gives +1.03 to +1.27% at 300 K against +0.81% or +1.04% from two disagreeing sources, and an ordered ice XI proxy gives the measured negative sign with a magnitude 4 to 11 times too large [43].

D. Polymorph pairs

Hydrogen-containing structures without carbon were taken from the Materials Project [44] (4421 entries, fetched with mp-api on 21 October 2026) and the JARVIS-DFT `dft_3d` snapshot of 24 September 2025 [45] (3436 entries). Formula equality uses the reduced elemental composition. Structures were reduced to primitive cells and deduplicated with pymatgen's StructureMatcher (length tolerance 0.2, site tolerance 0.15, angle tolerance 5°), with different space groups at a symmetry precision of 0.05 Å blocking a merge and Materials Project entries preferred. A pair is two distinct structures of one composition whose source energies (Materials Project corrected formation energies, or JARVIS OptB88vdW formation energies, never mixed) differ by at most 25 meV/atom and whose primitive cells hold at most 32 atoms. Polymorph a is the one with the lower database energy. This gives 836 structures and 965 pairs; 569 pairs have a screen result in at least one of the three models (572 counting SevenNet). Database energies are used only to select and orient pairs; all physics comes from the MLIP relaxations and phonons.

A structure has covalent X-H bonds when the nearest non-H neighbour of every H lies within 1.3 Å. Of the 965 pairs, 220 (63 formulas) have clean results for covalent X-H bonds in at least one model; these 521 pair-model rows (MACE 188, ORB 163, MatterSim 170) form the slope fit. They include 41 rows of pairs without any O-H donor (X = B, N, P or F). The sign test uses O-H pairs only, defined as pairs in which at least one structure has an O-H donor; in every headline structure all H are bonded to O, so the mean X-H length there is the mean O-H length.

E. Regime, sign rule, baselines and statistical unit

For each pair and model, $x = \bar{d}_{\mathrm{XH}}(b) - \bar{d}_{\mathrm{XH}}(a)$ is the difference of the mean model X-H lengths in mÅ and $y = \Delta\Delta F_{\mathrm{iso}}$ per H. The rule predicts $\mathrm{sign}(y) = \mathrm{sign}(x)$: deuteration favours the polymorph with the shorter mean O-H bond. The regime variable r_{max} is the larger of the two mean model O-H lengths. An evaluation (one pair computed with one model) enters the sign test when $|x| \geq 5$ mÅ. The strong side is $r_{\mathrm{max}} \geq 0.99$ Å.

The 0.99 Å boundary, the choice of r_{max}, the 5 mÅ eligibility and the O-H definition were set after inspecting the pre-freeze data, and the frozen result keeps them. We re-chose the boundary inside each leave-one-formula-out training fold (grid 0.97 to 1.03 Å in 0.01 Å steps, maximizing right minus wrong on the training strong side, lowest threshold winning ties) and scored the held-out formula; this covers the threshold value only. The alternatives for the other choices are shown in a multiverse table (Sec. III B). The baseline is a constant sign, the majority sign of y in each leave-one-formula-out training fold. Because a is the lower-energy database polymorph,

a positive constant sign means "deuteration destabilises the polymorph with the higher database energy". Competitor descriptors (density, MLIP static energy, O···O) are scored on the same rows, with their direction fixed physically and, separately, chosen in training folds. An orientation-invariant null flips the predicted sign of each pair (shared by its models) at random; 2×10^5 draws give its distribution.

Evaluations share structures, models and formulas. We therefore report pair majorities (majority over the models of a pair) and formula majorities alongside evaluation counts, give formula-clustered bootstrap intervals (2000 resamples of formulas with replacement) for slopes and accuracies, and add Theil-Sen slopes and a structure-level fit with formula fixed effects, in which the per-structure isotope term $F_{\mathrm{D}} - F_{\mathrm{H}}$ per H is regressed on mean X-H length within each formula, which removes pair combinatorics. These analyses were added in revision and are computed by `src/isoshift_event/paper_stats.py` from the frozen tables into `results/paper_stats.json`.

The counts were frozen at a fixed point (the compute freeze), after which the headline sentence was set by a written decision; later analyses are labelled post-freeze.

F. Validation against measured spectra

For every O-H donor in the clean structures we compared the harmonic Γ-point stretch frequency with the empirical correlation of Libowitzky [46], $\nu = 3592 - 304\times 10^9 \exp(-d_{\mathrm{O\cdots O}}/0.1321)$ cm^{-1} with d in Å, built from measured infrared and Raman spectra of minerals. The stretch mode of a donor is the Γ mode above 1500 cm^{-1} with the largest projection on that O-H bond, the acceptor is the nearest other O with H···O below 2.6 Å (no angle criterion), and symmetry-equivalent donors are merged. In each O···O window the model frequency is regressed on the Libowitzky frequency at the same distance; the slope ratio f equals 1 when the measured slope is reproduced. Intervals are 95% bootstrap intervals over structures (2000 resamples).

G. Prediction-first protocol

For each test against experiment the model predictions and scoring rule were committed to git before the measured values were read; the literature agents that collected measurements did not see the predictions, and the measured rows were committed before the comparison. Three qualifications are disclosed with the results: the scoring rule of the mineral fractionation test was committed together with its result; the free energies at 0, 600 and 2000 K existed before the temperature hypotheses were committed, so the temperature rescoring is retrospective even though the decomposition tests had fixed criteria; and one temperature test was scored after a code fix prompted by a failing run (Sec. III F). The prospective test fixed signs, eligibility and scoring before any of its phonon jobs ran. Commit identifiers are in Table II; the AI-agent workflow is described in the AI-use statement.

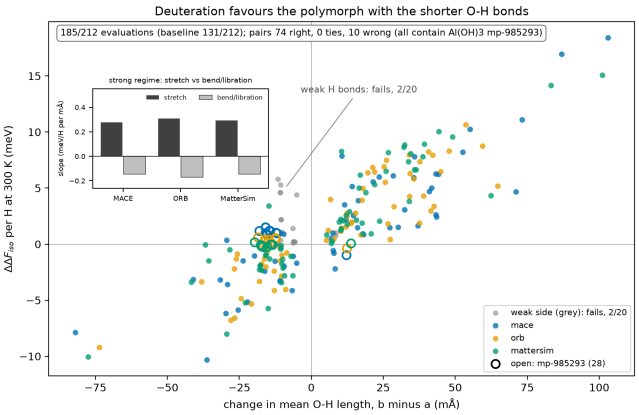


FIG. 1. The short-bond rule on the headline rows. Each point is one O-H pair computed with one model, with $|\Delta$ mean O-H$| \geq 5$ mÅ. Horizontal axis: change in mean model O-H length from a to b; vertical axis: $\Delta\Delta F_{\mathrm{iso}}$ per H at 300 K. Points in the upper right and lower left quadrants follow the rule. Coloured: $r_{\max} \geq 0.99$ Å; grey: both polymorphs below (weak hydrogen bonds). The boundary was chosen after inspecting the data. Open markers: the 28 evaluations involving Al(OH)$_3$ mp-985293, kept in every count. The inset shows strong-regime (O-H, $r_{\max} \geq 0.99$ Å) slopes of the stretch and of the bend and libration terms, without uncertainties. The figure title states the rule, not a measured result.

III. RESULTS

A. Slope of the isotope term against bond length

Across the 220 pairs $\Delta\Delta F_{\mathrm{iso}}$ per H at 300 K rises with the change of mean X-H length (Fig. 2, Table I). Least-squares slopes over all clean covalent rows are 0.112 (MACE), 0.117 (ORB) and 0.128 (MatterSim) meV per H per mÅ with Pearson r of 0.76, 0.83 and 0.84. The range 0.11 to 0.13 is the model spread. Formula-clustered bootstrap intervals are 0.068 to 0.154, 0.083 to 0.165 and 0.090 to 0.169. Theil-Sen slopes, less sensitive to the few pairs with $|x|$ up to about 100 mÅ, are 0.061, 0.111 and 0.109; the MACE slope depends on a few high-influence pairs, and Al(OH)$_3$ alone supplies 43 of its 188 rows. Leaving out one formula at a time moves the slopes within 0.096 to 0.124 (MACE), 0.111 to 0.135 (ORB) and 0.121 to 0.145 (MatterSim). Restricted to O-H pairs the slopes are 0.124, 0.135 and 0.145. The structure-level fit with formula fixed effects, which uses each structure once, gives 0.120 (0.072 to 0.160; 167 structures, 55 formulas), 0.116 (0.087 to 0.153) and 0.126 (0.091 to 0.163). A positive slope means that the polymorph with the longer X-H bonds, which in these solids is the one with the stronger

hydrogen bonds and the shorter donor-acceptor distance, is destabilised by deuteration. The slopes are associations across polymorph pairs, not derivatives obtained by stretching one structure. A difference of 10 mÅ shifts the H/D free-energy balance by about 1.1 to 1.3 meV per H; the median difference between Materials Project O-H polymorphs is 4.1 mÅ in database geometries.

The mode-resolved integral splits the slope into two opposing terms. Over all rows the stretch term has a slope of +0.245, +0.259 and +0.252 meV per H per mÅ and the bend and libration term −0.133, −0.142 and −0.124, so 0.49 to 0.55 of the stretch slope is cancelled. In the O-H strong regime the total slopes are 0.132, 0.136 and 0.147, the stretch slopes 0.277, 0.308 and 0.293 and the bend and libration slopes −0.145, −0.171 and −0.146, a compensation of 0.52, 0.56 and 0.50. The two populations differ and are quoted separately. SevenNet has 23 clean covalent rows and an all-row slope of 0.105; it is drawn in Fig. 2 for comparison and not used in any claim.

B. The sign rule

The frozen statement of the result is as follows. When at least one polymorph has a mean model O-H length of 0.99 Å or more (a model-derived boundary for moderate to strong hydrogen bonds, chosen after inspecting the data), deuteration favours the polymorph with the shorter O-H bonds in 185 of 212 O-H model evaluations with $|\Delta$ mean O-H$| \geq 5$ mÅ (84 pairs, 28 formulas), against 131 of 212 for a constant-sign baseline fitted on leave-one-formula-out training folds. By model majority 74 pairs are right and 10 are wrong, and all 10 wrong pairs contain one $Al(OH)_3$ structure, mp-985293.

The more conservative units tell the same story. By formula majority the rule is right in 27 of 28 formulas; the one wrong formula is $Al(OH)_3$, all of whose headline rows involve mp-985293. Weighting formulas equally the rule is right in 0.92 of rows (bootstrap interval 0.85 to 0.98) against 0.49 for the baseline. The pooled accuracy, 0.87, has a formula-clustered interval of 0.68 to 0.98. The 212 evaluations come from only 89 distinct structures, and five oxyhydroxide and hydroxide formulas (AlOOH 39 rows, FeOOH 31, $Al(OH)_3$ 28, MnOOH 16, CrOOH 15) contribute 129 of them, so the result is weighted towards these families. Per model the rule is right in 52 of 68 MACE, 64 of 73 ORB and 69 of 71 MatterSim evaluations. SevenNet, on its own 10 headline rows, is right in all 10 (baseline 7); as a fourth vote the count is 195 of 222 against 138, with 75 pairs right and the same 10 wrong.

Baselines and competitors. The constant-sign baseline (131 of 212) is the statement that deuteration destabilises the polymorph with the higher database energy. Against an orientation-invariant null that assigns each pair a random predicted sign, 185 lies far in the tail: the null has mean 106 and standard deviation 10.7, its largest value in 2×10^5 draws is 154, so $p < 10^{-5}$ (the draws are not independent across shared structures, so this p-value describes the pair-level arrangement, not a sampling probability). A density rule, "deuteration destabilises the denser polymorph", is right in 104 of 212, at chance; ordering by MLIP static energy gives 137. The O···O sign, "deuteration favours the polymorph with the longer mean donor O···O", is right in 183 of the 211 headline rows that have an acceptor, against 184 for the O-H rule on the same rows (pair majority 73 right, 1 tie, 10 wrong); the two differences are correlated ($r = -0.80$). On like-for-like rows the two descriptors are therefore equivalent in sign. The O···O version needs no phonons, which makes it the practical form for screening, but it still uses relaxed geometries; with database geometries the sign agreement is lower (Sec. V). Choosing each competitor's direction inside training folds leaves these counts unchanged, except for density (72 of 212, because its training-fold direction is unstable).

Where the rule is reliable. The rule's skill sits in rows far from zero (Fig. 3b). For $|\Delta\Delta F_{\mathrm{iso}}| \geq 1$ meV per H it is right in 151 of 159 rows; below 1 meV per H in 34 of 53, close to the baseline rate. Models disagree in sign in 19 of the 74 headline pairs computed with at least two models; the median spread across models within a pair is 1.36 meV per H against a median $|\Delta\Delta F_{\mathrm{iso}}|$ of 2.37. Normalised by that spread, the rule is right in all 81 rows with $|\Delta\Delta F_{\mathrm{iso}}|$ above twice the spread and in 42 of 43 between one and two spreads, but in 53 of 78 below one spread. The stored classical-limit flag, which fires when $|\Delta\Delta F_{\mathrm{iso}}(2000\ \mathrm{K})| > |\Delta\Delta F_{\mathrm{iso}}(600\ \mathrm{K})|$ and so marks pairs near a sign crossover in temperature, separates the outcome almost completely: unflagged rows are right in 168 of 170, flagged rows in 17 of 42, including all 28 mp-985293 rows. This flag is a diagnostic tied to the outcome and is not used as a filter. Where the predicted effect exceeds the model spread the rule is nearly always right; near balance it has little skill.

The boundary. Re-choosing the boundary inside each leave-one-formula-out training fold picks 1.00 Å in 30 of 31 folds and 0.99 Å only in the fold that holds out $Al(OH)_3$, and gives 183 of 210 held-out evaluations right against a baseline of 131; in that $Al(OH)_3$ fold the rule scores 8 of 28 against a baseline of 19. Coverage changes with the boundary (Fig. 3a): moving it from 0.98 to 1.02 Å raises accuracy from 185 of 229 to 114 of 114 while the scored rows fall by half, so accuracy and coverage must be read together, and 183 of 210 and 185 of 212 are not directly comparable. The other analysis choices matter as much (multiverse, `results/paper_stats.json`): with the regime defined by the smaller of the two mean O-H lengths the rule is right in 138 of 144 (baseline 100; pairs 56 right, none wrong), by their mean in 177 of 188 (114; pairs 74 right, 5 ties, none wrong), and by the shorter mean O···O ≤ 2.93 Å in 185 of 222 (141); with eligibility at 3 or 10 mÅ instead of 5 it is right in 203 of 232 (141) and 165 of 188 (120). The rule beats the baseline under every variant, but a sharp, universal

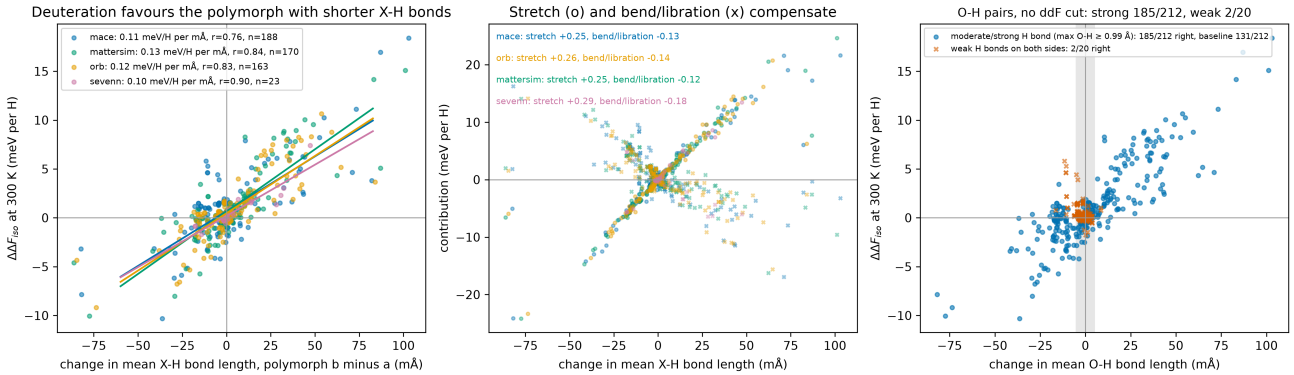


FIG. 2. All clean covalent rows of the screen at 300 K. Left: $\Delta\Delta F_{\mathrm{iso}}$ per H against the change in mean X-H length with per-model least-squares lines; legends give slope, r and the number of rows (SevenNet drawn for comparison only). Middle: mode-resolved stretch contribution (circles) and bend and libration contribution (crosses), with per-model slopes rounded to two decimals (three-decimal values in the text). Right: O-H pairs without a cut on $|\Delta\Delta F_{\mathrm{iso}}|$, against the change in mean O-H length: blue, at least one polymorph with mean model O-H ≥ 0.99 Å (185 of 212 eligible rows right, baseline 131); orange, both below (2 of 20 right). Rows inside the grey band, $|\Delta| < 5$ mÅ, are plotted but not scored.

TABLE I. Per-model slopes and counts. Slopes in meV per H per mÅ at 300 K. "All rows": all clean covalent rows of the screen; "strong": O-H rows with $r_{\max} \geq 0.99$ Å. Intervals are formula-clustered bootstrap 95% intervals. Compensation is the ratio of the bend and libration slope to the stretch slope. Sign counts use O-H pairs with $r_{\max} \geq 0.99$ Å and $|\Delta| \geq 5$ mÅ, no cut on $|\Delta\Delta F_{\mathrm{iso}}|$. Temperatures other than 300 K are post-freeze rescorings at fixed geometry. Sources: `results/decompose_pairs.csv`, `results/F01_cv.json`, `results/paper_stats.json`, `results/post_freeze/F09_temperature.json`.

	MACE	ORB	MatterSim	Pooled
Clean covalent rows (formulas)	188 (55)	163 (45)	170 (51)	521 (63)
All-row slope, OLS	0.112	0.117	0.128	
95% interval	0.068 to 0.154	0.083 to 0.165	0.090 to 0.169	
All-row slope, Theil-Sen	0.061	0.111	0.109	
Structure-level slope, formula fixed effects	0.120	0.116	0.126	
Pearson r (all rows)	0.76	0.83	0.84	
All-row stretch / bend and libration	+0.245 / −0.133	+0.259 / −0.142	+0.252 / −0.124	
All-row compensation	0.54	0.55	0.49	
Strong-regime rows in fit	97	97	99	
Strong-regime total slope	0.132	0.136	0.147	
Strong-regime stretch / bend and libration	0.277 / −0.145	0.308 / −0.171	0.293 / −0.146	
Strong-regime compensation	0.52	0.56	0.50	
Sign rule right, evaluations, 300 K	52/68	64/73	69/71	185/212
Constant-sign baseline, 300 K				131/212
Pair majority right/tie/wrong				74/0/10 of 84
Formula majority right				27 of 28
Sign rule right, 0 K	44/68	54/73	55/71	153/212
Sign rule right, 600 K	67/68	73/73	70/71	210/212
Stretch slope ratio f vs Libowitzky, 2.50 to 2.80 Å	0.91	0.92	0.92	
Isotope ZPE difference MAE vs DFT (meV per H)	1.5	0.7	1.3	

boundary is not established, and the boundary is an empirical region of these model geometries.

The Al(OH)$_3$ exception. Structure mp-985293 is a 28-atom $P2_12_12_1$ polymorph of Al(OH)$_3$ listed by the Materials Project as experimental, with no imaginary modes in any of the three models. Its mean model O-H of 0.996 (MACE), 0.997 (ORB) and 0.999 Å (MatterSim) sits just above the boundary, while its donors span 0.987 to 1.005 Å and its mean O$\cdots$O contacts, 2.77 to 2.82 Å, are shorter than those of its partners; it is denser, 6.79 to 6.94 Å^3/atom against 7.53 to 7.88 for its headline partners. In its 28 headline evaluations the stretch term has

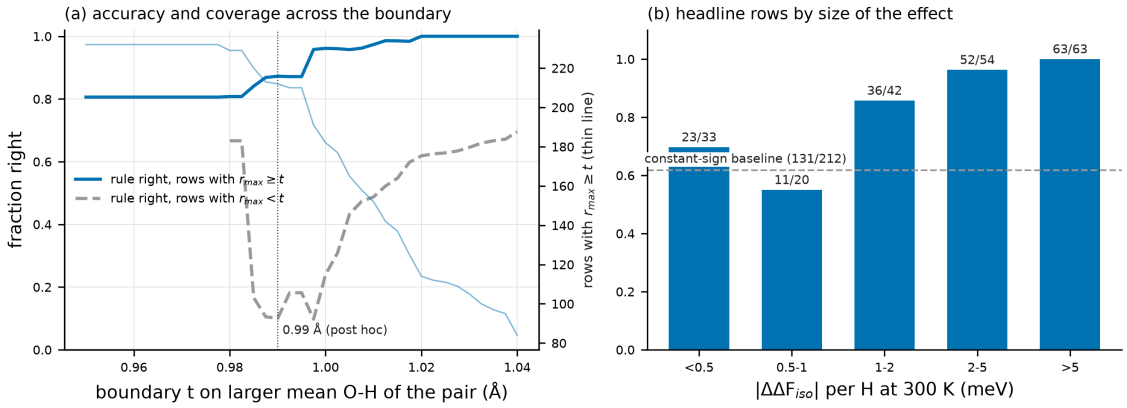


FIG. 3. Where the rule holds. (a) Fraction of O-H evaluations ($|\Delta| \geq 5$ mÅ, three models, 232 rows) for which the rule is right, on each side of a boundary t on $r_{\max}$, with the number of rows on the strong side (thin line, right axis). The 0.99 Å boundary (dotted) was chosen after inspecting the data. (b) Headline rows binned by $|\Delta\Delta F_{\rm iso}|$ per H at 300 K, with the number right over the number in the bin; dashed line, the constant-sign baseline rate. Data: `results/paper_stats.json`.

the predicted sign, but bend and libration cancel 95 to 132% of it instead of about half (MACE 1.19 to 1.32, ORB 1.02 to 1.14, MatterSim 0.95 to 1.02, against a median of 0.50 in the other strong-regime rows). MACE calls 0 of 10 of its pairs right, ORB 0 of 9 and MatterSim 8 of 9, with every MatterSim value at $|\Delta\Delta F_{\rm iso}| \leq 0.26$ meV per H. Here bond length alone stops predicting the sign. We do not claim a phase identity or a mechanism for its larger compensation; a post-freeze test of whether the mean O-H$\cdots$O angle predicts it failed (Table II). As post hoc locators chosen after seeing the frozen data, leaving out the 10 pairs with mp-985293 gives 177 of 184 and raising the boundary to 1.00 Å gives 175 of 182; these are nearly the same intervention, because 28 of the 30 evaluations between 0.99 and 1.00 Å come from mp-985293. Neither replaces the frozen count.

C. A physically cleaner subset

The clean criterion does not exclude unconverged relaxations or imaginary modes. Of the 212 headline rows, 119 carry a stored relaxation-not-converged flag and 161 a soft-mode flag (flags at pair level, kept from the screen). Using the per-structure records of the screen (relaxation converged to 10^{-4} eV/Å, fraction of imaginary modes on the H-mass mesh), the rule is right in 86 of 93 rows in which both relaxations converged (52 pairs: 45 right, 7 ties, none wrong; baseline 66), and in 35 of 36 rows in which both structures are also free of imaginary modes (28 pairs: 27 right, 1 wrong; 9 formulas; baseline 28). This strictest subset contains no mp-985293 row and almost no ORB row, because ORB relaxations rarely reached 10^{-4} eV/Å; its all-row slopes are 0.206 (MACE, 30 rows) and 0.146 (MatterSim, 24 rows). The subset is small and selected by numerical behaviour that may correlate with chemistry, so it supports the rule's sign but does not fix the slope. No structure was re-relaxed or followed along a soft mode for this revision.

D. Weak hydrogen bonds: a reversed sign

When both polymorphs have mean model O-H below 0.99 Å, the rule is right in only 2 of 20 evaluations: the sign is reversed in 18 (13 pairs, 1 right and 12 wrong by majority; 7 formulas, 6 of them wrong by majority: $Al(OH)_3$, $Be(OH)_2$, $Cd(OH)_2$, NaOH, RbOH, $Zn(OH)_2$). Over all weak-side O-H rows the slopes are negative in all three models (-0.27 for MACE, 76 rows; -0.058 for MatterSim; -0.014 for ORB), and in the prospective test (Sec. IV) the weak side was right in 2 of 9. This is a consistent reversal at modest confidence, and it matches the known change of sign of the net quantum effect with hydrogen-bond strength [16, 18]: for weak bonds the stretch barely changes with geometry (the Libowitzky curve spans only 170 $\rm cm^{-1}$ from 2.80 to 3.10 Å), and bend, libration and lattice terms set the sign. Which terms dominate was not resolved here, and the evidence is 20 evaluations in 7 formulas. Hydroxides such as NaOH, where a quantum paraelectric transition is driven by isotope and pressure [47], belong to this regime, outside the rule's domain.

E. The stretch term against measured spectra

For O-H donors with model O$\cdots$O distances of 2.50 to 2.80 Å the harmonic Γ-point stretch frequencies follow the slope of the Libowitzky correlation [46] with slope ratios of 0.91 (MACE, 52 donors in 18 structures), 0.92 (ORB, 58 in 20) and 0.92 (MatterSim, 58 in 17), with bootstrap intervals of 0.67 to 1.07, 0.82 to 1.01 and 0.81 to 1.02 (Fig. 4). The measured curve spans 1646 $\rm cm^{-1}$

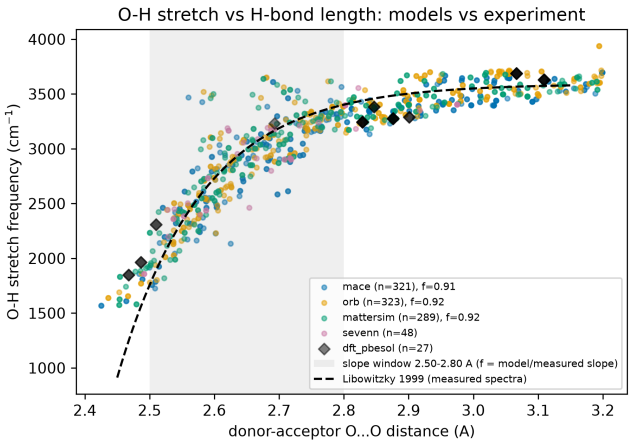


FIG. 4. Harmonic Γ-point O-H stretch frequencies of the model donors against donor-acceptor O···O distance (all donors; legend n counts all donors, while f is computed on 52 to 58 donors inside the shaded window), with the empirical correlation of Libowitzky built from measured spectra [46] (dashed) and PBEsol phonons of 27 donors (diamonds), only one of which lies inside the window. The shaded window, 2.50 to 2.80 Å, is where the slope ratio f (model over empirical slope) is evaluated. Donors far above the curve near 2.55 to 2.65 Å are not explained; no angle criterion is applied to the acceptor search. Harmonic model frequencies are compared with anharmonic measured fundamentals.

in this window. The model frequencies are harmonic and the measured ones fundamentals, and the gap between them, which matters for isotope partition functions [48], grows with hydrogen-bond strength, so the harmonic reference slope is flatter than the measured one; with an assumed growth of that gap by 150 to 400 cm^{-1} over the window, the corrected ratios are 1.00 to 1.22. The assumed range is not a confidence interval. If the bend and libration term were unaffected, the rule's sign would change only at ratios of about 1.9 to 2.3; the mp-985293 pairs show that the net sign is sensitive to that term. Above 2.80 Å the ratios are 0.67, 2.61 and 1.46, so ORB and MatterSim are too steep there, and the last 0.13 Å of the strong regime (to O···O of about 2.93 Å) is not covered. This is a consistency check on the stretch term, not a validation of isotope free energies or of the bend and libration term.

F. Temperature dependence

The isotope free energies stored before the freeze at 0, 300, 600 and 2000 K allow the same 212 evaluations to be rescored at fixed geometry. This rescoring is post hoc; its mechanism was then tested with four hypotheses committed before any new decomposition run. The rule is right in 153, 185 and 210 of 212 evaluations at 0, 300 and 600 K, against leave-one-formula-out baselines of 145, 131 and 114, and in 211 at 2000 K. At 0 K the rule barely beats the baseline; it is a thermal rule. At 0 K the unflagged rows are right in 153 of 170, so the misses there are again largely the crossover pairs. The mp-985293 rows score 0, 8 and 28 of 28 at 0, 300 and 600 K. All 212 rows pass the quadrature check at 0 and 300 K and 190 at 600 K (24 of the 28 mp-985293 rows); quadrature and direct $\Delta\Delta F_{\rm iso}$ have the same sign in all 212 rows at every temperature.

Of the four predeclared hypotheses three passed and one failed. The stretch term keeps the predicted sign in all 28 mp-985293 rows at 0 and 600 K, with compensation above 1 at 0 K and below 1 at 600 K (passed): these pairs cross over in temperature rather than through a change of the stretch sign. The crossover lies between 431 and 506 K for MACE, 317 and 406 K for ORB and mostly below 300 K for MatterSim (descriptive). The Teller-Redlich product rule holds at every q to 2.6×10^{-11} per H in these structures (passed). At 2000 K the sign of $\Delta\Delta F_{\rm iso}$ equals the sign predicted by the change in the trace of the on-site H force constants in 211 of 212 rows (bar 202, passed); this test initially scored 124 of 212 because the scoring script read the pair orientation from the wrong file, and it was rescored after that fix with hypothesis, threshold and rows unchanged. With the stretch near 5000 K, $\hbar\omega/k_{\rm B}T$ is still about 2.5 at 2000 K, so this is a sign test, not evidence that every mode is classical. The polymorph with the shorter mean O-H has the larger H force-constant trace in all 212 rows, so at high temperature the rule reduces to the Bigeleisen-Mayer and Urey statement that deuterium concentrates where the H force constants are largest [13, 14]. The hypothesis that failed was the proposed size of the temperature trend: compensation falls with temperature in 198 of 212 rows, but by about 39% at the median between 0 and 600 K (median C 0.665, 0.582 and 0.406 at 0, 300 and 600 K), and the predeclared drop of more than 40% in at least 80% of rows was met in only 98. Temperatures above about 500 K describe the harmonic isotope term at fixed geometry, not phase stability; gibbsite and many hydrates decompose below 600 K.

G. An analytic explanation within the same models

The following estimate was derived after the archived release and changes no frozen number. For one harmonic mode, $F = k_{\rm B}T\ln[2\sinh(\hbar\omega/2k_{\rm B}T)]$, and the isotope ratio of an O-H stretch is $\rho=\omega_{\rm H}/\omega_{\rm D}=\sqrt{\mu_{\rm D}/\mu_{\rm H}}$ with the reduced masses of ^{16}O with H and D, $\rho=1.373607$ and $1/\rho=0.728010$. Using $1/\sqrt{2}$ instead overstates the zero-point shift by about 8% and the high-temperature slope by about 6%. Let $s=-d\nu_{\rm str}/dr_{\rm OH}$ be the softening of the stretch wavenumber as the O-H bond lengthens. Linearising $\Delta\Delta F_{\rm iso}$ in the bond-length difference gives the stretch slope

$$S_{\rm str}=\frac{hc}{2}\,s\left[\coth\frac{x}{2}-\frac{1}{\rho}\coth\frac{x}{2\rho}\right]\simeq\frac{hc}{2}\left(1-\frac{1}{\rho}\right)s, \quad (4)$$

with $x = hc\nu_{\mathrm{H}}/k_{\mathrm{B}}T$. For stretch wavenumbers of 3000 to 3600 cm^{-1} the bracket equals its zero-temperature value to within 1.5×10^{-4} (relative) at 300 K, so $S_{\mathrm{str}} = 0.016861\,s$ meV per H per mÅ with s in cm^{-1} per mÅ. A longer O-H bond softens the stretch, which makes $F_{\mathrm{D}} - F_{\mathrm{H}}$ less negative, so from the stretch alone deuteration favours the polymorph with the shorter O-H bond.

The input s is taken from the screen's own phonons (donors with stretch weight above 0.5 and mean O-H of 0.96 to 1.05 Å): the stretch frequencies fall by 19.0 (MACE), 19.3 (ORB) and 19.5 (MatterSim) cm^{-1} per mÅ, with $r = -0.99$ to -1.00 (17 to 21 over narrower windows). The comparison is therefore partly circular: s and the projected stretch term come from the same phonons, and an experimental $d\nu/dr_{\mathrm{OH}}$ was not checked against a primary source and is not used. With $s = 19$ to 19.5, Eq. (4) gives $S_{\mathrm{str}} = 0.32$ to 0.33 meV per H per mÅ, against fitted stretch slopes of 0.245 to 0.259 (all rows) and 0.277 to 0.308 (strong O-H regime); the one-mode estimate is 4 to 34% high, and these fitted values come from two populations. Two likely reasons, not tested, are that the regressor is the mean O-H length of all H in the cell while s is fitted per donor, and that the stretch is not a pure mode. The estimate thus checks the projection and linearisation within the models; it is an explanation, not an independent validation.

Bends and librations harden as the hydrogen bond strengthens. With $s_b = +d\nu_b/dr_{\mathrm{OH}}$ for each of $n_b = 2$ bending modes per H, $S_{\mathrm{bend}} = -n_b s_b (hc/2)[\coth(x_b/2) - \rho_b^{-1}\coth(x_b/2\rho_b)]$, and at 0 K with $\rho_b \approx \rho$ the compensation is $C = n_b s_b/s$. The observed C of about 0.5 requires $s_b \approx s/4$, roughly 5 cm^{-1} per mÅ per bend; this value is inferred from the compensation, not measured. Relative to 0 K the stretch weight is 1.000 at 300 K and 0.988 at 600 K, while a 1200 cm^{-1} bend falls to 0.941 and 0.687 and a 900 cm^{-1} libration to 0.858 and 0.556. With $s = 18$, $s_b = 4.5$ and $\nu_b = 1400$ cm^{-1}, C falls from 0.50 at 0 K to 0.48 at 300 K and 0.38 at 600 K. This agrees in direction with the temperature rescoring, but it does not explain the rise from 153 to 185 right between 0 and 300 K; that would need librations and lattice modes below about 900 cm^{-1} and rows near zero, a plausibility argument, not a derivation. The assumptions are harmonic modes, an isolated O-H stretch (anharmonicity, which lowers the measured O-H/O-D ratio to about 1.33 to 1.36, is outside the estimate), linear dependence of frequencies on O-H length and bends treated as two modes of one frequency.

H. Exploratory prediction for diaspore and δ-AlOOH

Diaspore (α-AlOOH, *Pnma*, mp-625152) transforms at high pressure to δ-AlOOH ($P2_1nm$, mp-23807), which has the stronger hydrogen bond: its O-H is longer by 24 to 32 mÅ and its O···O shorter by 60 to 90 mÅ at 0 GPa in the models. The rule therefore predicts that deuteration destabilises δ-AlOOH. At 0 GPa geometry the harmonic isotope term $\Delta\Delta F_{\mathrm{iso}}(\delta - \text{diaspore})$ at 300 K is +0.52, +0.77 and +0.83 meV/atom (MatterSim, MACE, ORB), with $\Delta V = -0.30$ to -0.32 Å^3/atom, which gives +0.3 to +0.4 GPa. Quasi-harmonic Gibbs energies at 10, 15 and 20 GPa, with one linear fit for both isotopes, place the 300 K transition of AlOOD 1.33 GPa (MACE) and 1.13 GPa (ORB) above that of AlOOH; a temperature-resolved rebuild gives +1.33, +1.13 and +0.99 GPa (MatterSim) at 300 K and +0.94, +0.81 and +0.73 GPa at 1000 K. The quasi-harmonic isotope term at 10, 15 and 20 GPa, 2.30, 2.42 and 2.22 meV/atom (MACE) and 1.86, 2.30 and 2.50 (ORB), is about three times its 0 GPa harmonic value, which is why the pressure shift exceeds the 0 GPa estimate. Its growth with pressure is consistent with the δ-phase hydrogen bond strengthening towards symmetrization, exactly where harmonic physics is least reliable; O-H and O···O along the pressure grid were not stored and are not reported.

The models place the AlOOH transition at 8.76 (MACE, extrapolated below the 10 GPa grid point), 13.00 (ORB) and 15.14 GPa (MatterSim) at 300 K, 4 to 6 GPa apart; none of these values was compared with the experimental stability field [49] or with earlier DFT work [50], and the experimental boundary is crossed at high temperature. δ-AlOOH undergoes isotope-dependent proton disorder in the same pressure window [1, 2], which harmonic physics omits; the measured 4.4 GPa shift of that neighbouring transition shows that anharmonic isotope effects in this system are large, and the potentials overbind hydrogen bonds (Sec. IV). The prediction is therefore an exploratory, conditional sign prediction outside the rule's validated domain; neither its sign nor its size is validated. A logged search found no measurement or calculation of the isotope shift of this boundary. Related measured shifts point the same way, the δ-AlOOH disorder transition [1] and ε-FeOOD remaining asymmetric above the pressure at which ε-FeOOH symmetrises [51], but both are different transitions. The same calculation for goethite to ε-FeOOH gives +0.2 to +0.5 GPa with a mesh sensitivity as large as the signal, so only its sign is stated.

IV. TESTS AGAINST EXPERIMENT

No polymorph isotope free energy has been measured, so none of the tests below scores the rule directly. Each tests an ingredient or a consequence (Table II).

Volume isotope effects (F02, mixed). Quasi-harmonic volume isotope effects were predicted for known H compounds before the measured cell volumes were read. α-AlH_3 has the measured sign [52] with a magnitude 1.3 to 1.5 times too large; GaOOH and β-CrOOH have the wrong sign against cross-laboratory references of limited comparability. The misses lie in the strong hydrogen-bond regime, where quasi-harmonic physics omits Ubbelohde-type proton anharmonicity [3]. The

TABLE II. Every prediction-first, blind or prospective test, with the commit that fixed the prediction before its data or result. "Model" tests compare with code checks, not with measurements.

Test	What was predicted	Commit	Outcome	Notes
P01/F02	Volume isotope effects of known H compounds	0fffc54	Mixed	α-AlH_3 right sign, 1.3 to 1.5 times too large; GaOOH and β-CrOOH wrong sign against references of limited quality. Does not test the rule.
P02/F04	Stretch and bend shifts, goethite to ε-FeOOH and diaspore to δ-AlOOH	da560cf	Bend sign agrees, stretch sign not determined	Goethite to ε-FeOOH: bend rises as predicted; the stretch comparator range (-511 to -11 cm^{-1}) does not fix the stretch sign. Diaspore pair inconclusive.
P03/F04	Stretch and bend shifts, manganite to groutite	c990f11	Inconclusive	
P04/F05	Harmonic mineral D/H fractionation, 118 structures	f3ac0ac	Ordering only	Signs 6 of 6 and order brucite > goethite > manganite right; brucite minus goethite 5 to 10 times too large; scoring rule committed with its result (90ad4cd).
P05/F07	H_2O to D_2O shift of three ice boundaries	bec68d7	Failed	Ih-II and Ih-III wrong sign in 3 of 3 models; II-III right sign; secondary data.
P06/F08	Rule refined with the O-H$\cdots$O angle	cdb32da	Failed	No gain over the bare rule; all 10 wrong pairs stay wrong.
P08/F10	Structure-only atlas, 1796 pairs	d991991	Nothing scorable	0 of 26 literature rows usable.
P09/F09	Temperature hypotheses H1 to H4	d03f8d2	H1 failed, H2 to H4 passed	H4 passed after a disclosed orientation fix (initial run 124 of 212).
P10/F11	Prospective test on pairs never computed	eb61ccf	Failed	16 of 111 right; 87 rows lost to predeclared numerical rules; on 24 clean rows 16 right, precommitted baseline 5, opposite constant sign 19; pair majority 5/1/4.
P11/F12	Boehmite against diaspore D/H fractionation	184fbee	Inconclusive	Sign agrees at 380 °C in 3 of 3 models; no stated uncertainty, diaspore 25% exchanged.
F06	Reimplementation without the original code	n/a	Code agrees	9 of 9 referenced rows to 10^{-9} meV/atom; tests code, not physics.

test passed where the rule has no skill and missed where it applies, so it does not test the rule; it suggests that the potentials overweight the O-H stretch contribution to volume derivatives in strongly hydrogen-bonded crystals.

Differential spectroscopy (F04, bend sign only). The mechanism predicts that, between two polymorphs, the stretch falls and the bend rises as the hydrogen bond strengthens. For goethite to ε-FeOOH the highest O-H bend rises by $+290$ cm^{-1} against $+195$ predicted [53]. The stretch comparator is the midpoint of the broad 2600 to 3100 cm^{-1} band reported for ε-FeOOH, giving a nominal -260 cm^{-1} against -273 predicted but anywhere from -511 to -11 cm^{-1}; Mashino et al. [54] read the same spectrum as two maxima near 2800 to 2900 and 3300 to 3400 cm^{-1}, and if the upper component were the main band the stretch sign would reverse, so the stretch sign is not determined. The goethite bands are attributed to Liu et al. [55], whose full text was not read. The diaspore to δ-AlOOH and manganite to groutite pairs were inconclusive. This is consistent with the sign of the compensating bend shift at the zone centre, not a measurement of its size; the measured bend to stretch shift ratio, about 1.1, exceeds the model median of 0.7.

Mineral D/H fractionation (F05, ordering only). The same isotope free energies give reduced partition function ratios, so differences between measured mineral-water fractionation factors test them without a model of water. Predictions for 118 structures were committed before the data were read. The measured factors are attributed in our data file to Xu and Zheng for brucite [56], Yapp for goethite [57] and Hariya and Tsutsumi for manganite [58], each reached through secondary sources and not checked against the primary papers. The sign of every scored difference (6 of 6, in 2 scored pairs) and the order brucite > goethite > manganite in D affinity are right in all three models, and the mineral with the longer O-H bonds takes up less deuterium. The magnitudes are not: for brucite minus goethite the models give 190 to 300 per mil against a measured 22 to 46, 5 to 10 times too large, and the predicted difference falls with temperature while the measured one rises. The order is assembled from two temperature windows, and the factor of 5 to 10 applies to brucite minus goethite; the two pairs with manganite give ratios of 1.6 and 0.7, one of them unscored. Corrections to harmonic reduced partition function ratios are

discussed in Ref. [48]. Brucite (mean O-H 0.971 Å) lies outside the rule's regime. The scoring rule was committed together with its result.

Ice phase boundaries (F07, failed). The H_2O to D_2O shifts of three ice boundaries were predicted before any measured D_2O boundary was read. For Ih-II the models give −12 to −140 MPa at 230 to 250 K against a measured triple-point shift of +12 to +15 MPa; Ih-III is also wrong in sign in all three models, while II-III is right in sign and within a factor of 2 to 3. The measured values come from a secondary compilation and from Bridgman's D_2O work via a reference equation of state [59, 60], with an inferred isotope label and no stated uncertainty. Both failures involve ice Ih, modelled through the ordered ice XI proxy, 7 to 9% too dense in all three models; overbound hydrogen bonds and the omitted proton disorder are both consistent with the failure.

Prospective test (F11, failed). After the freeze we committed signs for 111 primary evaluations of pairs never computed before (42 pairs, 20 formulas, both phases experimental), after relaxation and before any phonon run. The bar was at least 75% right, at least 15 points above the committed constant-sign baseline and a pair majority right in at least 80% of pairs. The rule got 16 of 111 right, the baseline 5. 87 rows were lost to the predeclared numerical rules (69 failed or dynamically unstable, 18 not clean), dominated by KH_2PO_4-type and ice polymorphs with imaginary modes, which capped the attainable accuracy at 21.6%. The surviving rows do not rescue the rule: on the 24 clean rows it was right in 16, the pre-committed baseline in 5 and the opposite constant sign, chosen after the fact, in 19, and the pair majority was 5 right, 1 tie and 4 wrong. The failure combines numerical attrition with insufficient skill on the rows that survived. Transfer of the rule to new pairs is not shown, and the clean filter selects a population, ordered oxyhydroxides and hydroxides, close to the one on which the in-sample rule was built. Descriptively and after part of the data had been seen, the clean rows gave a total slope of 0.181 and a compensation of 0.44. The predeclared secondary test, at 600 K without a threshold, missed its 90% bar (21 of 120); the 21 rows it scored right are all of its clean rows.

Boehmite against diaspore (F12, inconclusive). Among the headline pairs only boehmite against diaspore has a measured route. The models predict that D prefers boehmite, the shorter O-H phase: +42.5, +30.3 and +41.3 per mil at 380 °C. Measured mineral-water fractionation [61], read from a compilation [62], gives a boehmite minus diaspore difference of +28.8 per mil at 380 °C (boehmite at 2 kbar, diaspore at 4 kbar), the predicted sign in all three models, but the source states no uncertainty and the diaspore value comes from 25% exchange, so the predeclared rule scores the case inconclusive.

Reimplementation (F06). A second model, without access to our code, rewrote the double difference and its split from the cached force constants with its own mesh and quadrature. Of the 20 rows it computed, 9 have a reference row; on these it reproduces $\Delta\Delta F_{\mathrm{iso}}$ to 10^{-9} meV/atom with 9 of 9 signs. The other 11 belong to pairs excluded as not clean. Both codes read the same force constants, so this checks implementation, not the potentials or the harmonic model.

V. DISCUSSION

Known physics and the new part. The competition between stretch and bend zero-point energies in hydrogen bonds is established [16–19], and the high-temperature direction, in which deuterium prefers the phase with the larger H force-constant trace, is the Bigeleisen-Mayer and Urey argument [13, 14]; at 600 K and above our rule reduces to it (211 of 212 at 2000 K). Hydrogen-bond control of mineral D/H fractionation is known qualitatively [20, 21], and isotopic polymorphism is documented in single systems [5–12]. The direction of the rule therefore follows from known physics, and Sec. III G reproduces the size of the stretch slope within the same models. What this work adds is narrower: the net slope of the polymorph isotope free-energy difference against O-H length across many inorganic polymorph pairs, with clustered uncertainties; the room-temperature outcome of the stretch against bend and libration competition, including where it is reliable (effects larger than the model spread) and where it reverses (weak hydrogen bonds); one structure in which bend and libration outweigh the stretch at room temperature but not above a crossover temperature; and a conditional pressure prediction. Within these models, the same pair can change its isotope preference with temperature at fixed geometry, and a bond-length rule cannot stand for the whole vibrational free energy. A longer O-H bond goes with a stronger hydrogen bond and, in strongly bonded systems, deuteration lengthens the O···O distance (the Ubbelohde effect [3, 22]); in chemical terms the rule says that deuteration favours the polymorph whose hydrogen bonds are weaker, the direction seen in the pentachlorophenol and 4-methylpyridine complex [6, 8]. Because the O···O sign performs as well as the O-H sign on the same rows, the practical form is a geometric screen; with database geometries instead of relaxed ones the sign agreement drops (69% against 86% in a pre-freeze analysis), so relaxed geometries are needed.

Limitations. (1) The free energies are harmonic or quasi-harmonic. Anharmonicity of the O-H stretch, tunnelling and proton disorder are omitted, and D is evaluated at the H geometry, so geometric isotope effects on O-H and O···O are not included; the failed or inconclusive tests (F05 magnitudes, F07, the δ-AlOOH disorder window) point to exactly these effects in strongly hydrogen-bonded systems, where the rule is meant to apply. (2) The three potentials share PBE-level training data and overbind hydrogen bonds (ice XI 7 to 9% too dense), so a shared bias in O-H lengths would move both axes of

Fig. 1 together; the per-structure benchmark errors (0.5 to 1.1 meV per H at 300 K) are comparable to the typical signal, and no DFT double difference was computed. (3) The regime boundary, regime variable, eligibility cut and O-H definition were set after inspecting the data; nested re-choice covers only the threshold value, and the multiverse shows that the count depends on these choices, although the rule beats the baseline under all of them. (4) Evaluations share structures and models; five formulas supply most rows, and the formula-clustered intervals are wide. (5) The clean criterion is narrower than physical stability; the strictest subset (converged, no imaginary modes) is small, and the prospective test on new pairs failed, through numerical losses and on the clean rows. (6) No polymorph isotope free energy has been measured.

The direct experiment. D/H partitioning between two coexisting polymorphs constrains $\Delta\Delta G_{\mathrm{iso}}$. For polymorphs a and b of one composition with n_{H} hydrogens per formula unit,

$$\Delta\Delta G_{\mathrm{iso}} = n_{\mathrm{H}} k_{\mathrm{B}} T \ln \alpha_{a\text{-}b} + R, \tag{5}$$

where $\alpha_{a\text{-}b}$ is the trace (dilute D) fractionation factor and R accounts for the difference between trace and full substitution; with a full-substitution factor the relation is exact with $R = 0$. To second order $R \approx (n_{\mathrm{H}} k_{\mathrm{B}} T/2)[\mathrm{Var}_b(l) - \mathrm{Var}_a(l)] + N_b - N_a$, with $l_i = \ln \beta_i$ per H site and N_p the non-additivity of full substitution. R vanishes when the H sites are equivalent and independent with composition-independent substitution free energies and ideal mixing; harmonicity alone does not establish this, because the rule of the geometric mean needs its own assumptions [63]. Both model phases of diaspore and δ-AlOOH have one inequivalent H site, so the variance term vanishes, but the non-additivity term has not been bounded, and a trace measurement constrains the predicted full-substitution difference only once it is. Equation (5) is the solid-state analogue of the Bigeleisen relation between vapour-pressure isotope effects and fractionation [14], and the boundary shift follows as $\Delta P_{\mathrm{iso}} \approx -\Delta\Delta G_{\mathrm{iso}}/\Delta V$ to linear order. As a conditional conversion with $R = 0$, the quasi-harmonic isotope term at each model's transition pressure corresponds to $1000 \ln \alpha(\mathrm{diaspore} - \delta) = 265$ to 363 per mil at 300 K and 66 to 90 per mil at 1000 K (`results/paper_stats.json`); these are not fractionation predictions, and the 5 to 10 times overestimate of harmonic mineral fractionation in F05 suggests that measured values would be smaller. Coexisting polymorphs exist only along the boundary at high pressure, and D/H exchange between solids needs a fluid and high temperature, where α is smaller; the feasibility of resolving such values with ion-probe D/H measurements was not assessed here. The second test is a matched, reversed-bracket determination of the diaspore to δ-AlOOH boundary for AlOOH and AlOOD in the same press, pressure medium, calibrant and temperature path, with each boundary approached from both phase assemblages. A shift of opposite sign contradicts the prediction; a shift not resolvable from zero is a null result. On the computational side, path-integral mass integration [23, 24] or self-consistent anharmonic phonons [64] on a few headline pairs, and DFT double differences on a stratified subset, are the controls this work lacks.

VI. CONCLUSION

In harmonic calculations with three universal ML potentials over 220 inorganic polymorph pairs, the H/D free-energy difference between polymorphs rises with the change in mean X-H bond length, 0.11 to 0.13 meV per H per mÅ across models with formula-clustered intervals of about 0.07 to 0.17, a stretch term about half cancelled by bend and libration. In crystals with moderate to strong hydrogen bonds deuteration favours the polymorph with the shorter O-H bonds in 74 of 84 pairs and 27 of 28 formulas (185 of 212 evaluations against 131 for a constant sign); the rule is nearly always right where the effect exceeds the model spread, has little skill near balance, reverses for weak hydrogen bonds and fails in one $Al(OH)_3$ structure where bend and libration outweigh the stretch at room temperature. It did not transfer in a prospective test: 16 of 111 committed rows were right, and on the 24 clean rows 16 against 19 for the opposite constant sign, with pair majority 5 right, 1 tie and 4 wrong. A one-mode harmonic estimate within the same models explains the sign and, within 4 to 34%, the size of the stretch slope, but not the rise in skill from 0 to 300 K. Exploratory quasi-harmonic calculations put the diaspore to δ-AlOOH transition of AlOOD about 1 GPa above AlOOH, outside the rule's validated domain. Anharmonic and DFT controls on a subset of pairs, D/H partitioning between coexisting polymorphs and matched AlOOH and AlOOD boundary runs would test the result.

DATA AND CODE AVAILABILITY

Code, data, prediction files, decision records and findings are at `https://github.com/canerganis/isoshift` [34]; this manuscript corresponds to branch `post-hackathon` at the commit that adds it. The version submitted to the OSI Hackathon 2026 is archived as v1.0.0 at Zenodo, doi:10.5281/zenodo.23578412 [65]. The headline counts are recomputed on a CPU in seconds from the frozen pair table with `python -m src.isoshift_event.rule_cv`, and the revision statistics with `src/isoshift_event/paper_stats.py`. The post-freeze analyses (F09, F11, F12) were committed on 22 October 2026, before the DOI was added to the repository; the analytic estimate (note M03) and the revision statistics were committed later and are not part of v1.0.0. Force constants and raw phonon outputs are not in the repository. Derived Materials Project data are CC BY 4.0; the licence of the redistributed

JARVIS-DFT structures was not verified.

ACKNOWLEDGMENTS

The author thanks the organisers of the OSI Hackathon 2026, Ben Blaiszik and colleagues, for the event in which this work was done; Valency Systems for access to the Valency Bond literature service; and the marimo team for the notebook framework used for the results application.

AI-USE STATEMENT

AI agents did the work under the author's direction, each with a fixed role; the author set the question, approved each step and made the calls that needed a person. (1) Conductor: Claude Opus 5.5 planned the work, ran the GPU screen with MACE, ORB, MatterSim and SevenNet, wrote the analysis and kept a ledger of every step. (2) Literature and blind-test agents: Claude Sonnet instances queried Valency Bond, each with one task (a validation value, an ice boundary, a fractionation factor); the blind-test agents never saw the predictions. All 182 queries made before submission are logged with their purpose in the repository. (3) Math checks: every formula was derived twice, independently, by GPT-6 Astra and Claude Opus, and the code used it only after the two agreed. (4) Councils: hard calls (the freeze, what to claim, what to drop) went to a council of Claude Opus, GPT-6 Astra and Gemini 4 Argon. Each member answered blind, then read and challenged the others; the chair (Claude Opus) decided by argument, not vote, and recorded the dissent. (5) Independent check: GPT-6 Astra, through Codex and without our code, rewrote the calculation from the equations and matched $\Delta\Delta F_{\mathrm{iso}}$ to 10^{-9} meV/atom, which checks the code, not the physics. After submission, Claude Haiku agents ran the two prior-art search rounds, and the analytic estimate was derived independently by GPT-6 Astra and Claude Opus and cross-checked numerically. This manuscript was drafted from the repository records by Claude Opus 5.5, which merged in parts of an independent draft by GPT-6 Astra and revised it after referee reports by GPT-6 Astra and Claude Opus. Agreement between agents is not treated as experimental validation. The author is responsible for the content.

[1] A. Sano-Furukawa, T. Hattori, K. Komatsu, H. Kagi, T. Nagai, J. J. Molaison, A. M. dos Santos, and C. A. Tulk, Direct observation of symmetrization of hydrogen bond in δ-AlOOH under mantle conditions using neutron diffraction, Sci. Rep. **8**, 15520 (2018).

[2] A. Sano-Furukawa, K. Komatsu, C. B. Vanpeteghem, and E. Ohtani, Neutron diffraction study of δ-AlOOD at high pressure and its implication for symmetrization of the hydrogen bond, Am. Mineral. **93**, 1558 (2008).

[3] A. R. Ubbelohde and K. J. Gallagher, Acid-base effects in hydrogen bonds in crystals, Acta Crystallogr. **8**, 71 (1955).

[4] S. Koval, J. Kohanoff, R. L. Migoni, and E. Tosatti, Ferroelectricity and isotope effects in hydrogen-bonded KDP crystals, Phys. Rev. Lett. **89**, 187602 (2002).

[5] S. Crawford, M. T. Kirchner, D. Bläser, R. Boese, W. I. F. David, A. Dawson, A. Gehrke, R. M. Ibberson, W. G. Marshall, S. Parsons, and O. Yamamuro, Isotopic polymorphism in pyridine, Angew. Chem. Int. Ed. **48**, 755 (2009).

[6] J. Zhou, Y.-S. Kye, and G. S. Harbison, Isotopomeric polymorphism, J. Am. Chem. Soc. **126**, 8392 (2004).

[7] J. Zhou, Y.-S. Kye, A. I. Kolesnikov, and G. S. Harbison, Origins of isotopomeric polymorphism, Isotopes Environ. Health Stud. **42**, 271 (2006).

[8] B. C. K. Ip, I. G. Shenderovich, P. M. Tolstoy, J. Frydel, G. S. Denisov, G. Buntkowsky, and H.-H. Limbach, NMR studies of solid pentachlorophenol-4-methylpyridine complexes exhibiting strong OHN hydrogen bonds: geometric H/D isotope effects and hydrogen bond coupling cause isotopic polymorphism, J. Phys. Chem. A **116**, 11370 (2012).

[9] J. Falk, D. Hofmann, and K. Merz, Controlled usage of H/D exchange to circumvent concomitant polymorphs of ROY, IUCrJ **5**, 569 (2018).

[10] C. Shi, X. Zhang, C.-H. Yu, Y.-F. Yao, and W. Zhang, Geometric isotope effect of deuteration in a hydrogen-bonded host-guest crystal, Nat. Commun. **9**, 481 (2018).

[11] S. J. Ford, O. J. Delamore, J. S. O. Evans, G. J. McIntyre, M. R. Johnson, and I. Radosavljević Evans, Giant deuteron migration during the isosymmetric phase transition in deuterated 3,5-pyridinedicarboxylic acid, Chem. Eur. J. **17**, 14942 (2011).

[12] K. Merz and A. Kupka, Deuterium perturbs the molecular arrangement in the solid state, Cryst. Growth Des. **15**, 1553 (2015).

[13] H. C. Urey, The thermodynamic properties of isotopic substances, J. Chem. Soc. , 562 (1947).

[14] J. Bigeleisen and M. G. Mayer, Calculation of equilibrium constants for isotopic exchange reactions, J. Chem. Phys. **15**, 261 (1947).

[15] R. M. Badger, A relation between internuclear distances and bond force constants, J. Chem. Phys. **2**, 128 (1934).

[16] R. H. McKenzie, C. Bekker, B. Athokpam, and S. G. Ramesh, Effect of quantum nuclear motion on hydrogen bonding, J. Chem. Phys. **140**, 174508 (2014).

[17] R. H. McKenzie, B. Athokpam, and S. G. Ramesh, Isotopic fractionation in proteins as a measure of hydrogen bond length, J. Chem. Phys. **143**, 044309 (2015).

[18] X.-Z. Li, B. Walker, and A. Michaelides, Quantum nature of the hydrogen bond, Proc. Natl. Acad. Sci. U.S.A. **108**, 6369 (2011).

[19] T. E. Markland and B. J. Berne, Unraveling quantum mechanical effects in water using isotopic fractionation, Proc. Natl. Acad. Sci. U.S.A. **109**, 7988 (2012).

[20] T. Suzuoki and S. Epstein, Hydrogen isotope fractionation between OH-bearing minerals and water, Geochim. Cosmochim. Acta **40**, 1229 (1976).

[21] M. Méheut, M. Lazzeri, E. Balan, and F. Mauri, First-principles calculation of H/D isotopic fractionation between hydrous minerals and water, Geochim. Cosmochim. Acta **74**, 3874 (2010).

[22] H.-H. Limbach, P. M. Tolstoy, N. Pérez-Hernández, J. Guo, I. G. Shenderovich, and G. S. Denisov, OHO hydrogen bond geometries and NMR chemical shifts: from equilibrium structures to geometric H/D isotope effects, with applications for water, protonated water, and compressed ice, Isr. J. Chem. **49**, 199 (2009).

[23] M. Ceriotti and T. E. Markland, Efficient methods and practical guidelines for simulating isotope effects, J. Chem. Phys. **138**, 014112 (2013).

[24] B. Cheng and M. Ceriotti, Direct path integral estimators for isotope fractionation ratios, J. Chem. Phys. **141**, 244112 (2014).

[25] M. Ceriotti, W. Fang, P. G. Kusalik, R. H. McKenzie, A. Michaelides, M. A. Morales, and T. E. Markland, Nuclear quantum effects in water and aqueous systems: experiment, theory, and current challenges, Chem. Rev. **116**, 7529 (2016).

[26] T. E. Markland and M. Ceriotti, Nuclear quantum effects enter the mainstream, Nat. Rev. Chem. **2**, 0109 (2018).

[27] M.-Z. Du, S.-Y. He, J. Shen, Z. Ye, J.-X. Zeng, D. H. Zhang, V. Kapil, and W. Fang, Isotope effects at classical cost through mass-differentiable machine learning (2026), arXiv:2610.11325.

[28] M. Rossi, P. Gasparotto, and M. Ceriotti, Anharmonic and quantum fluctuations in molecular crystals: a first-principles study of the stability of paracetamol, Phys. Rev. Lett. **117**, 115702 (2016).

[29] V. Kapil, E. Engel, M. Rossi, and M. Ceriotti, Assessment of approximate methods for anharmonic free energies, J. Chem. Theory Comput. **15**, 5845 (2019).

[30] V. Kapil and E. A. Engel, A complete description of thermodynamic stabilities of molecular crystals, Proc. Natl. Acad. Sci. U.S.A. **119**, e2111769119 (2022).

[31] N. Bourgeois, J.-C. Crivello, P. Cenedese, V. Paul-Boncour, and J.-M. Joubert, Vibration analysis of hydrogen, deuterium and tritium in metals: consequences on the isotope effect, J. Phys.: Condens. Matter **30**, 335402 (2018).

[32] H. Lee, V. I. Hegde, C. Wolverton, and Y. Xia, Accelerating high-throughput phonon calculations via machine learning universal potentials, Mater. Today Phys. **53**, 101688 (2025).

[33] A. Loew, D. Sun, H.-C. Wang, S. Botti, and M. A. L. Marques, Universal machine learning interatomic potentials are ready for phonons, npj Comput. Mater. **11**, 178 (2025).

[34] M. C. Erganiş, IsoShift source code and data, `https://github.com/canerganis/isoshift` (2026).

[35] I. Batatia, P. Benner, Y. Chiang, A. M. Elena, D. P. Kovács, J. Riebesell, *et al.*, A foundation model for atomistic materials chemistry (2023), arXiv:2401.00096.

[36] B. Rhodes, S. Vandenhaute, V. Šimkus, J. Gin, J. Godwin, T. Duignan, and M. Neumann, Orb-v3: atomistic simulation at scale (2025), arXiv:2504.06231.

[37] H. Yang, C. Hu, Y. Zhou, X. Liu, Y. Shi, J. Li, G. Li, Z. Chen, S. Chen, C. Zeni, M. Horton, R. Pinsler, *et al.*, MatterSim: A deep learning atomistic model across elements, temperatures and pressures (2024), arXiv:2405.04967.

[38] Y. Park, J. Kim, S. Hwang, and S. Han, Scalable parallel algorithm for graph neural network interatomic potentials in molecular dynamics simulations, J. Chem. Theory Comput. **20**, 4857 (2024).

[39] J. Kim, J. Kim, J. Kim, J. Lee, Y. Park, Y. Kang, and S. Han, Data-efficient multi-fidelity training for high-fidelity machine learning interatomic potentials (2024), arXiv:2409.07947.

[40] A. Togo and I. Tanaka, First principles phonon calculations in materials science, Scr. Mater. **108**, 1 (2015).

[41] A. Togo, L. Chaput, T. Tadano, and I. Tanaka, Implementation strategies in phonopy and phono3py, J. Phys.: Condens. Matter **35**, 353001 (2023).

[42] B. C. Chakoumakos, J. Horita, and V. O. Garlea, H/D isotope effects in brucite at low temperatures, Am. Mineral. **98**, 1 (2013).

[43] B. Pamuk, J. M. Soler, R. Ramírez, C. P. Herrero, P. W. Stephens, P. B. Allen, and M.-V. Fernández-Serra, Anomalous nuclear quantum effects in ice, Phys. Rev. Lett. **108**, 193003 (2012).

[44] A. Jain, S. P. Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, and K. A. Persson, Commentary: The Materials Project: A materials genome approach to accelerating materials innovation, APL Mater. **1**, 011002 (2013).

[45] K. Choudhary, K. F. Garrity, A. C. E. Reid, B. DeCost, A. J. Biacchi, A. R. Hight Walker, Z. Trautt, J. Hattrick-Simpers, A. G. Kusne, A. Centrone, *et al.*, The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design, npj Comput. Mater. **6**, 173 (2020).

[46] E. Libowitzky, Correlation of O-H stretching frequencies and O-H···O hydrogen bond lengths in minerals, Monatsh. Chem. **130**, 1047 (1999).

[47] S. Schaack, E. Mangaud, E. Fallacara, S. Huppert, P. Depondt, and F. Finocchi, When quantum fluctuations meet structural instabilities: the isotope- and pressure-induced phase transition in the quantum paraelectric NaOH, Phys. Rev. Lett. **131**, 126101 (2023).

[48] Q. Liu, J. A. Tossell, and Y. Liu, On the proper use of the Bigeleisen-Mayer equation and corrections to it in the calculation of isotopic fractionation equilibrium constants, Geochim. Cosmochim. Acta **74**, 6965 (2010).

[49] E. Ohtani, K. Litasov, A. Suzuki, and T. Kondo, Stability field of new hydrous phase, δ-AlOOH, with implications for water transport into the deep mantle, Geophys. Res. Lett. **28**, 3991 (2001).

[50] J. Tsuchiya, T. Tsuchiya, S. Tsuneyuki, and T. Yamanaka, First principles calculation of a high-pressure hydrous phase, δ-AlOOH, Geophys. Res. Lett. **29**, 10.1029/2002GL015417 (2002).

[51] O. Ikeda, H. Yamamoto, A. Sano-Furukawa, T. Sakamaki, T. Kuribayashi, Y. Noda, and A. Suzuki, High-pressure neutron diffraction study on ε-FeOOH: the spin-reorientation transition and hydrogen-bond symmetrization, J. Am. Chem. Soc. **147**, 4005 (2025).

[52] J. W. Turley and H. W. Rinn, Crystal structure of aluminum hydride, Inorg. Chem. **8**, 18 (1969).

[53] E. C. Thompson, A. H. Davis, N. M. Brauser, Z. Liu, V. B. Prakapenka, and A. J. Campbell, Phase transitions in ε-FeOOH at high pressure and ambient temperature, Am. Mineral. **105**, 1769 (2020).

[54] I. Mashino, S. Yamashita, and T. Yoshino, High-pressure spectroscopic investigation of ε-FeOOH: toward a better understanding of pressure-induced hydrogen-bond symmetrization, Phys. Chem. Miner. **53**, 18 (2026).

[55] H. Liu, T. Chen, C. Qing, Q. Xie, and R. L. Frost, Confirmation of the assignment of vibrations of goethite: an ATR and IES study of goethite structure, Spectrochim. Acta A **116**, 154 (2013).

[56] B.-L. Xu and Y.-F. Zheng, Experimental studies of oxygen and hydrogen isotope fractionations between precipitated brucite and water at low temperatures, Geochim. Cosmochim. Acta **63**, 2009 (1999).

[57] C. J. Yapp, Oxygen and hydrogen isotope variations among goethites (α-FeOOH) and the determination of paleotemperatures, Geochim. Cosmochim. Acta **51**, 355 (1987).

[58] Y. Hariya and M. Tsutsumi, Hydrogen-isotopic composition of some hydrous manganese minerals, Chem. Geol. **34**, 43 (1981).

[59] P. W. Bridgman, The pressure-volume-temperature relations of the liquid, and the phase diagram of heavy water, J. Chem. Phys. **3**, 597 (1935).

[60] S. Herrig, M. Thol, A. H. Harvey, and E. W. Lemmon, A reference equation of state for heavy water, J. Phys. Chem. Ref. Data **47**, 043102 (2018).

[61] C. M. Graham, S. M. F. Sheppard, and T. H. E. Heaton, Experimental hydrogen isotope studies. I. Systematics of hydrogen isotope fractionation in the systems epidote-H_2O, zoisite-H_2O and AlO(OH)-H_2O, Geochim. Cosmochim. Acta **44**, 353 (1980).

[62] T. Chacko, D. R. Cole, and J. Horita, Equilibrium oxygen, hydrogen and carbon isotope fractionation factors applicable to geologic systems, Rev. Mineral. Geochem. **43**, 1 (2001).

[63] W. J. Albery and M. H. Davies, Effect of the breakdown of the rule of the geometric mean on fractionation factor theory, Trans. Faraday Soc. **65**, 1059 (1969).

[64] L. Monacelli, R. Bianco, M. Cherubini, M. Calandra, I. Errea, and F. Mauri, The stochastic self-consistent harmonic approximation: calculating vibrational properties of materials with full quantum and anharmonic effects, J. Phys.: Condens. Matter **33**, 363001 (2021).

[65] M. C. Erganiş, IsoShift: the short-bond rule for deuteration and polymorph stability in hydrogen-bonded crystals, version 1.0.0, Zenodo (2026).